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FACULTAD DE INGENIERÍA DE SISTEMAS E INFORMÁTICA
DOCTORAL PROGRAM IN DEEP TECH, ARTIFICIAL INTELLIGENCE & EMERGING
TECHNOLOGIES



RESEARCH METHODS AND SCIENTIFIC INTEGRITY IN AI AND ADVANCED
TECHNOLOGIES

Deliverable:

PRISMA 2020 Flow Diagram

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TITLE: Explainable Artificial Intelligence for the Early Prediction of School Interruption Risk among Secondary Education Students in Peru

1. Proposed research topic

The proposed doctoral research focuses on the development and evaluation of an Explainable Artificial Intelligence model for the early identification of secondary school students at risk of interrupting their studies in Peru.

The systematic literature review is intended to identify existing Artificial Intelligence and Machine Learning approaches used for school dropout prediction and to determine the methodological, technological, and contextual gaps that could justify the proposed research.

2. Objective of the systematic literature review

The objective of this systematic literature review is to identify, analyze, and synthesize empirical studies that have applied Artificial Intelligence, Machine Learning, predictive analytics, or related computational techniques to predict or identify the risk of school interruption or dropout among secondary education students.

In addition, the review seeks to examine how existing studies have evaluated predictive performance, explainability, fairness, temporal validity, and educational usefulness.

3. Research Questions

- **SLR-RQ1**

How have Artificial Intelligence and Machine Learning approaches been used to predict or identify the risk of school interruption or dropout among secondary education students?

Spanish interpretation:

¿Cómo se han utilizado enfoques de Inteligencia Artificial y Machine Learning para predecir o identificar el riesgo de interrupción de estudios o abandono escolar en estudiantes de educación secundaria?

- **SLR-RQ2**

How have these studies evaluated predictive performance, calibration, temporal validity, explainability, explanation faithfulness and stability, fairness, and educational usefulness?

Spanish interpretation:

¿Cómo han evaluado estos estudios el desempeño predictivo, la calibración, la validez temporal, la explicabilidad, la fidelidad y estabilidad de las explicaciones, la equidad algorítmica y la utilidad educativa?

- **SLR-RQ3**

What methodological, technological, and contextual gaps remain in the existing evidence, particularly for Peru and comparable Latin American educational contexts?

Spanish interpretation:

¿Qué brechas metodológicas, tecnológicas y contextuales permanecen en la evidencia existente, particularmente para el Perú y contextos educativos latinoamericanos comparables?

4. Information sources

The literature search was conducted using the following information sources:

Source	Records identified
Scopus	104
ScienceDirect	382
ProQuest	46
Total	532

5. Search strategy

The search strategy was structured around three main conceptual blocks:

Concept 1 – School dropout

"school dropout" OR "student dropout" OR "early school leaving" OR "school interruption" OR "educational interruption" OR "school abandonment" OR "dropout risk" OR "student attrition" OR "school disengagement"

Concept 2 – Artificial Intelligence / Machine Learning

"artificial intelligence" OR "machine learning" OR "deep learning" OR "predictive analytics" OR "learning analytics" OR "early warning system"

Concept 3 – Secondary education

"secondary education" OR "secondary school*" OR "high school*" OR "lower secondary" OR "upper secondary" OR "K-12"

The general search logic was therefore:

Concept 1 AND Concept 2 AND Concept 3

The syntax was adapted when necessary to the limitations and characteristics of each information source.

6. Eligibility criteria

Studies were considered potentially eligible when they:

- addressed school dropout, interruption, early school leaving, or related outcomes;
- included secondary education students or a clearly separable secondary-school population;
- applied AI, Machine Learning, predictive analytics, learning analytics, or similar methods;
- developed or evaluated an approach for predicting or identifying students at risk;
- used real educational or student data;
- provided sufficient methodological information for analysis;
- were full peer-reviewed journal articles or eligible conference papers.

Studies were excluded when they:

- focused exclusively on higher education, universities, MOOCs, or unrelated populations;
- investigated educational outcomes different from dropout/interruption;
- did not apply an AI/ML or predictive approach;
- used AI for purposes unrelated to dropout prediction;
- were editorials, commentaries, abstracts, or other non-empirical publications;
- were systematic reviews, scoping reviews, bibliometric studies, or meta-analyses rather than primary studies;
- combined populations in a manner that prevented secondary education results from being separated.

7. Study selection process

A total of 532 records were initially identified.

After deduplication across and within information sources, **79 duplicate records were removed**.

Therefore:

$$532 - 79 = 453$$

A total of **453 unique records** entered title and abstract screening.

During this screening stage, **406 records were excluded** because they did not meet the predefined eligibility criteria.

Therefore:

$$453 - 406 = 47$$

A total of **47 reports were selected for full-text retrieval and eligibility assessment**.

At the time this assignment was prepared:

- 29 full-text reports had been assessed;

- 6 had been excluded after full-text assessment;
- 23 remained provisionally eligible;
- 18 were still pending full-text assessment.

8. PRISMA 2020 Flow Diagram

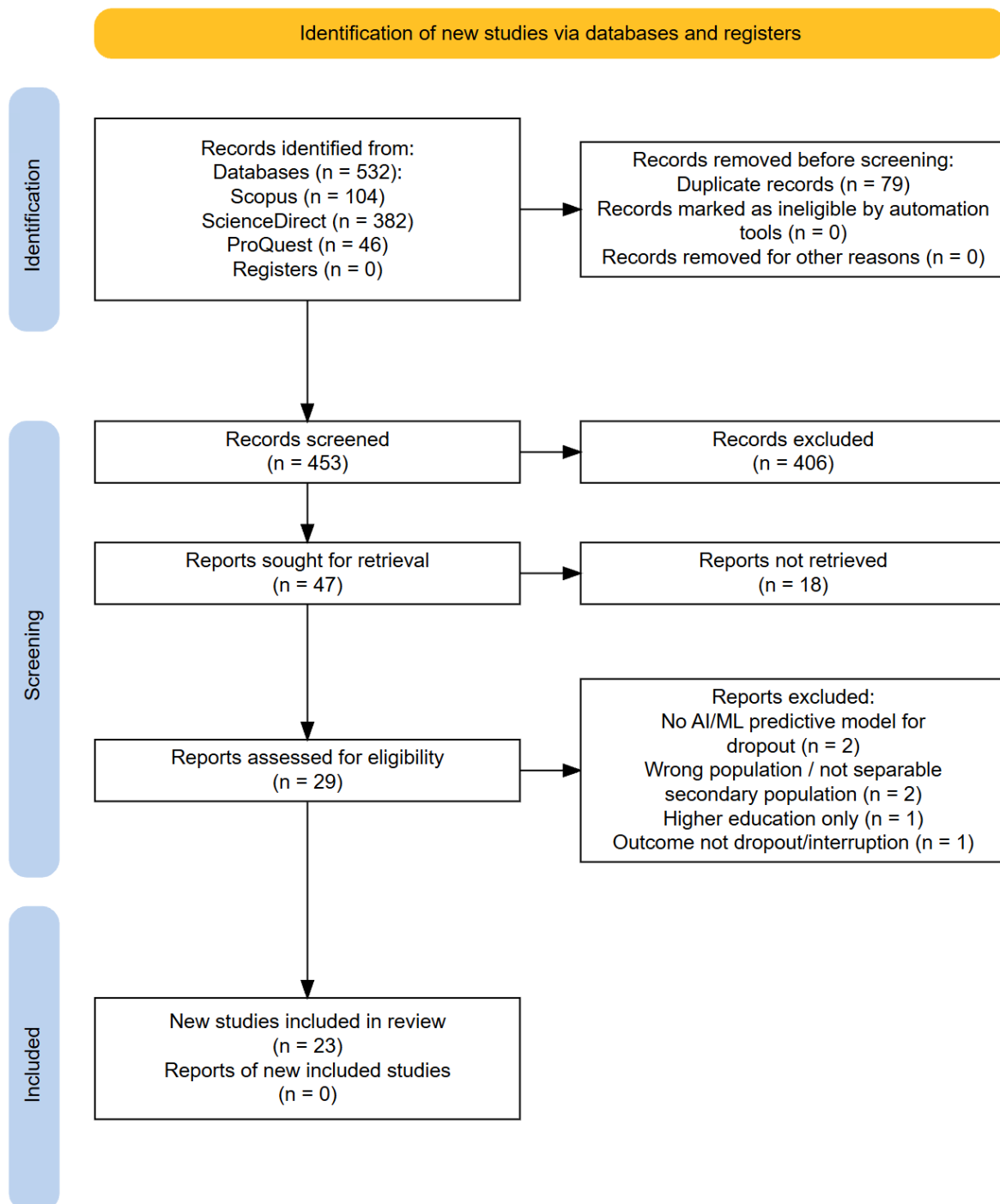


Figure 1. PRISMA 2020 flow diagram of the systematic literature review. Provisional status as of October 2, 2026.

9. Interpretation of the provisional PRISMA diagram

The systematic search identified **532 records** from Scopus, ScienceDirect, and ProQuest.

After removing **79 duplicates**, **453 unique records** were screened based on their titles and abstracts. Of these, **406 were excluded**, leaving **47 potentially eligible reports** for full-text assessment.

At the time of preparation of this PRISMA diagram, **29 reports had undergone full-text eligibility assessment**. Six reports were excluded because they did not fully satisfy the review criteria, while **23 studies remained provisionally eligible**.

The remaining **18 reports were still undergoing retrieval or full-text assessment**.

Therefore, the number of studies finally included in the systematic literature review should be considered **provisional rather than final**.

10. Preliminary observations

The preliminary screening indicates that the literature already contains studies that apply Machine Learning to school dropout prediction in secondary education.

In addition, some studies address specific aspects such as:

- Explainable Artificial Intelligence;
- feature importance;
- algorithmic fairness;
- class imbalance;
- early warning systems;
- human-machine collaboration;
- longitudinal prediction;
- temporal robustness and concept drift.

Consequently, the potential research gap should not be formulated simply as the absence of AI or Explainable AI models for school dropout prediction.

Instead, the emerging gap may involve the lack of an **integrated and rigorous evaluation combining predictive performance, calibration, explainability, explanation faithfulness and stability, fairness, temporal**

validation, and human-centered educational usefulness, particularly in the context of Peruvian secondary education.

This preliminary conclusion will be refined once the full-text eligibility assessment is completed.